

NOTEBOOK

Demara Williams DSW9740

Robotics NW16

Problem 1

$$l = 2 \text{ m}$$

$$\text{D.O.F} = 1$$

$$m = 1.5 \text{ kg}$$

6 constraints :-

$$q(0) = \dot{q}(0) = \ddot{q}(0) = 0$$

$$q(t_f) = q_f \quad \dot{q}(t_f) = \ddot{q}(t_f) = 0$$

Since there are 6 constraints, the trajectory can be modelled by a five (5) degree polynomial.

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

$$\dot{q}(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4$$

$$\ddot{q}(t) = 2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3$$

$$q(0) = 0 \Rightarrow a_0 = 0$$

$$\dot{q}(0) = 0 \Rightarrow a_1 = 0$$

$$\ddot{q}(0) = 0 \Rightarrow a_2 = 0$$

$$q(t) = a_3 t^3 + a_4 t^4 + a_5 t^5$$

$$\dot{q}(t) = 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4$$

$$\ddot{q}(t) = 6a_3 t + 12a_4 t^2 + 20a_5 t^3$$

$$q(t_f) = a_3 t_f^3 + a_4 t_f^4 + a_5 t_f^5 = q_f$$

$$\dot{q}(t_f) = 3a_3 t_f^2 + 4a_4 t_f^3 + 5a_5 t_f^4 = 0$$

$$\ddot{q}(t_f) = 6a_3 t_f + 12a_4 t_f^2 + 20a_5 t_f^3 = 0$$

Evaluating the system of 3 equations.

$$a_3 = \frac{10q_f}{t_f^3} \quad a_4 = -\frac{15q_f}{t_f^4} \quad a_5 = -\frac{6q_f}{t_f^5}$$

$\therefore$ Polynomial for the trajectory

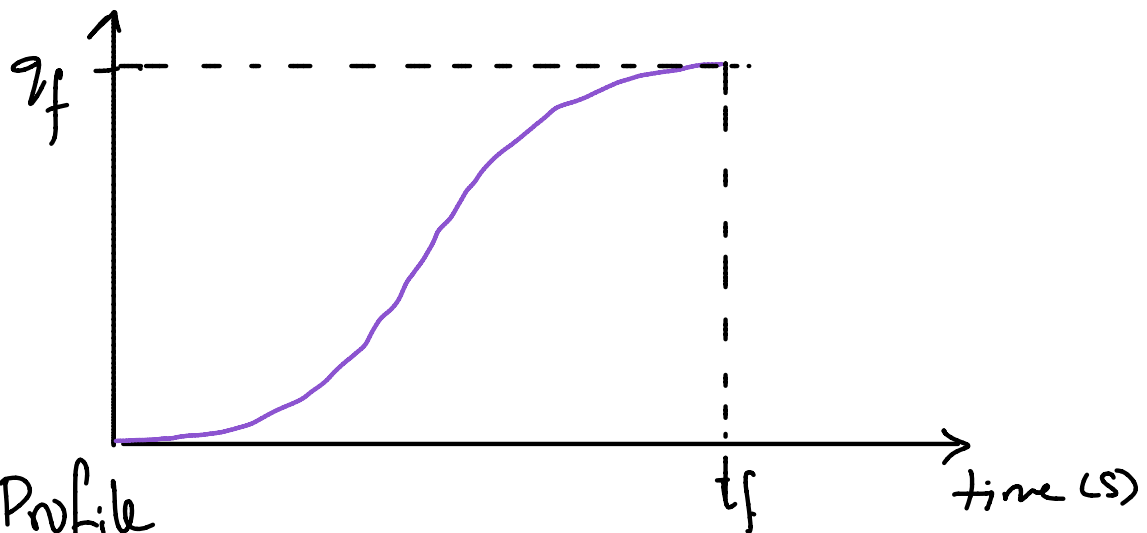
$$q(t) = 10q_f \left(\frac{t}{t_f}\right)^3 - 15q_f \left(\frac{t}{t_f}\right)^4 + 6q_f \left(\frac{t}{t_f}\right)^5$$

$$\dot{q}(t) = 30 q_f \frac{t^2}{t_f^3} - 60 q_f \frac{t^3}{t_f^4} + 30 q_f \frac{t^4}{t_f^5}$$

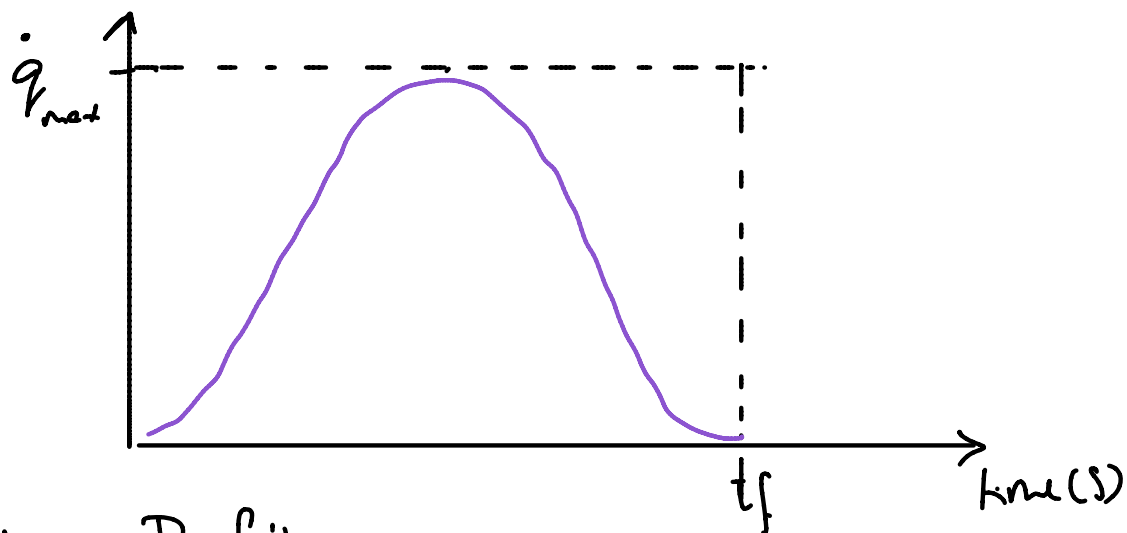
$$\ddot{q}(t_f) = 60 q_f \frac{t}{t_f^3} - 180 q_f \frac{t^2}{t_f^4} + 120 q_f \frac{t^3}{t_f^5}$$

for $t \in [0, t_f]$

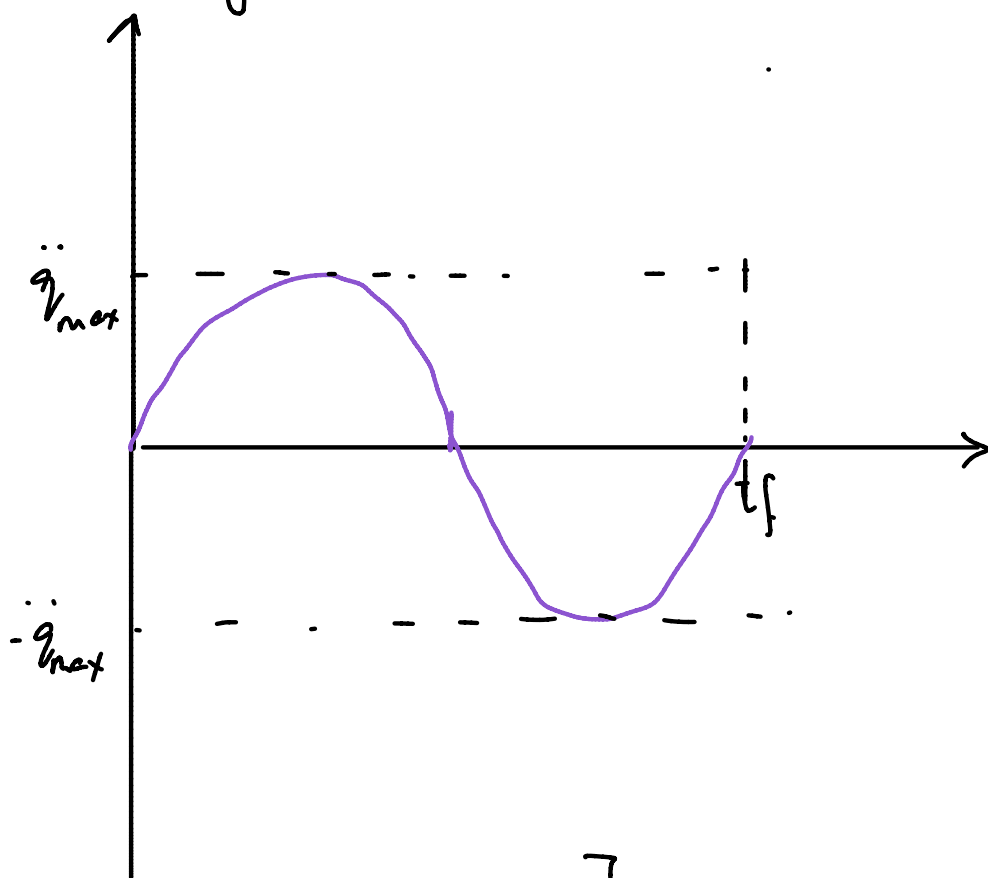
Position profile



Velocity Profile

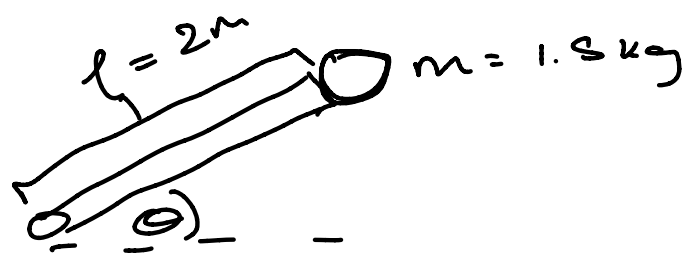


Acceleration Profile



$\forall t \in [0, t_f]$

1-DOF Manipulator



$$q_f = 180^\circ$$

$$= \pi$$

$$t_f = 0.75 \text{ sec}$$

$$q(t) = \frac{10\pi}{(0.75)^3} t^3 - \frac{15\pi}{(0.75)^4} t^4 + \frac{6\pi}{(0.75)^5} t^5$$

$$\therefore \ddot{q}(t) = \frac{60\pi}{(0.75)^3} t - \frac{180\pi}{(0.75)^4} t^2 + \frac{120\pi}{(0.75)^5} t^3$$

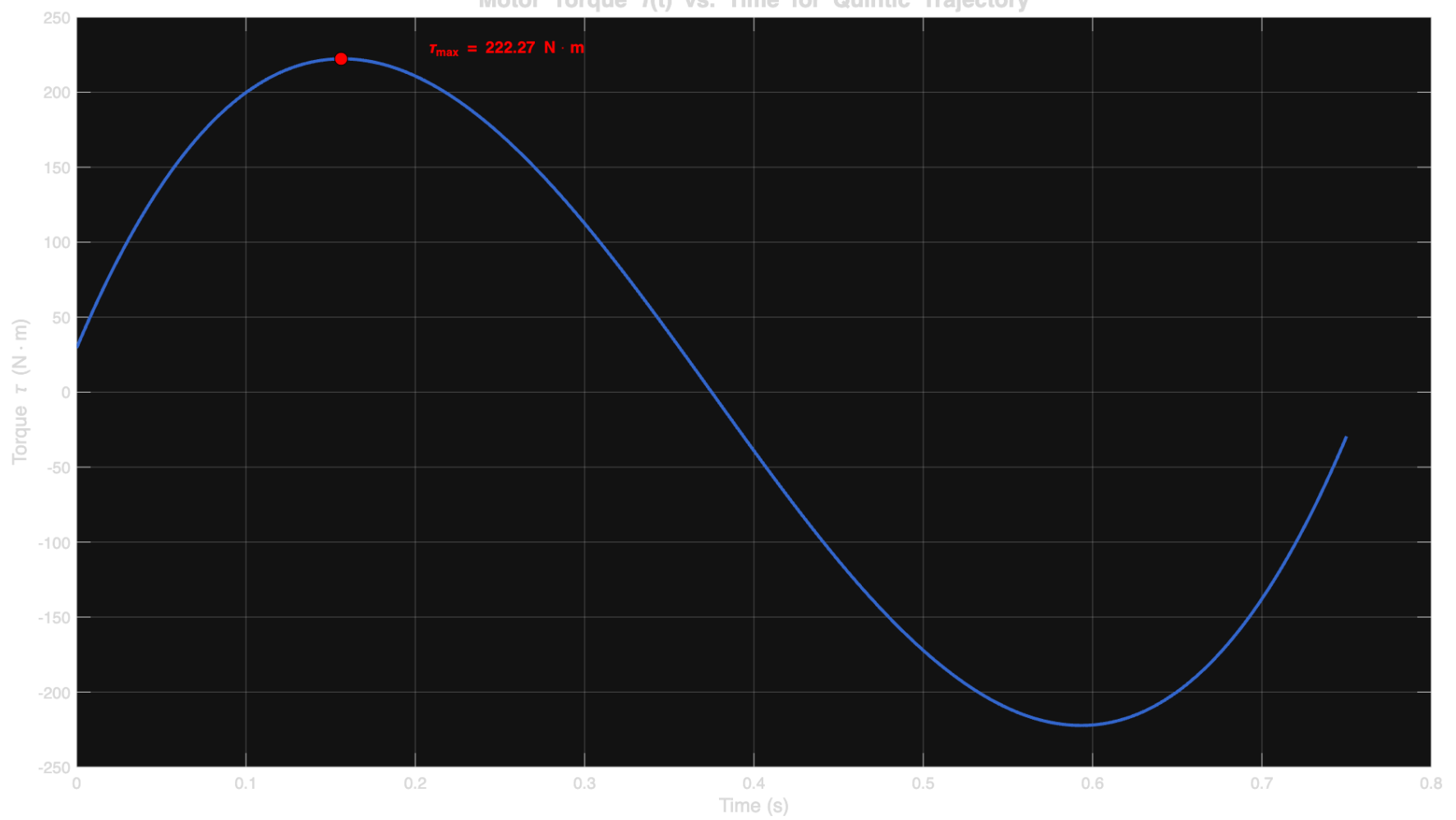
$$\tau = I \ddot{q} + mgl \cos(q)$$

$$I = ml^2 = 1.5 \times 2^2 = 6 \text{ kg} \cdot \text{m}^2$$

$$\therefore \tau(t) = 6 \ddot{q} + (1.5)(9.81)(2) \cos(q(t))$$

$$= 6 \ddot{q}(t) + 29.43 \cos(q(t))$$

Motor Torque $\tau(t)$ vs. Time for Quintic Trajectory



Based on the above plot of $\tau(t)$.

The maximum torque required

$$\tau_{\max} = 222.27 \text{ Nm}$$

Problem 2

(a). Set A

$G \rightarrow K$ is not possible

Set B

$M \rightarrow R$ is not possible

Set C

$M \rightarrow T$ is not possible

$\therefore$ Set D is the only set with transitions without collision.

(b). $R \rightarrow T$

$S \rightarrow G$

